CEREALBOX AGENCY · AI AUTOMATION & CREATIVE WORKFLOW PROPOSAL

One connected system, nine workflows, humans at every gate

Prepared by **Sohaib Ahmed** · November 2026

Every area below runs the same way: structured inputs, AI drafts, automated quality checks, and a person approving anything that reaches a client or an ad account. Approved work and live performance feed the Idea Generator, so each campaign starts with what the last one taught us.

☐ AI step ☐ Automated check / system step ☐ Human review or decision ☐ Data / asset

INTERVIEW CHECKLIST

What CerealBox asked us to prepare

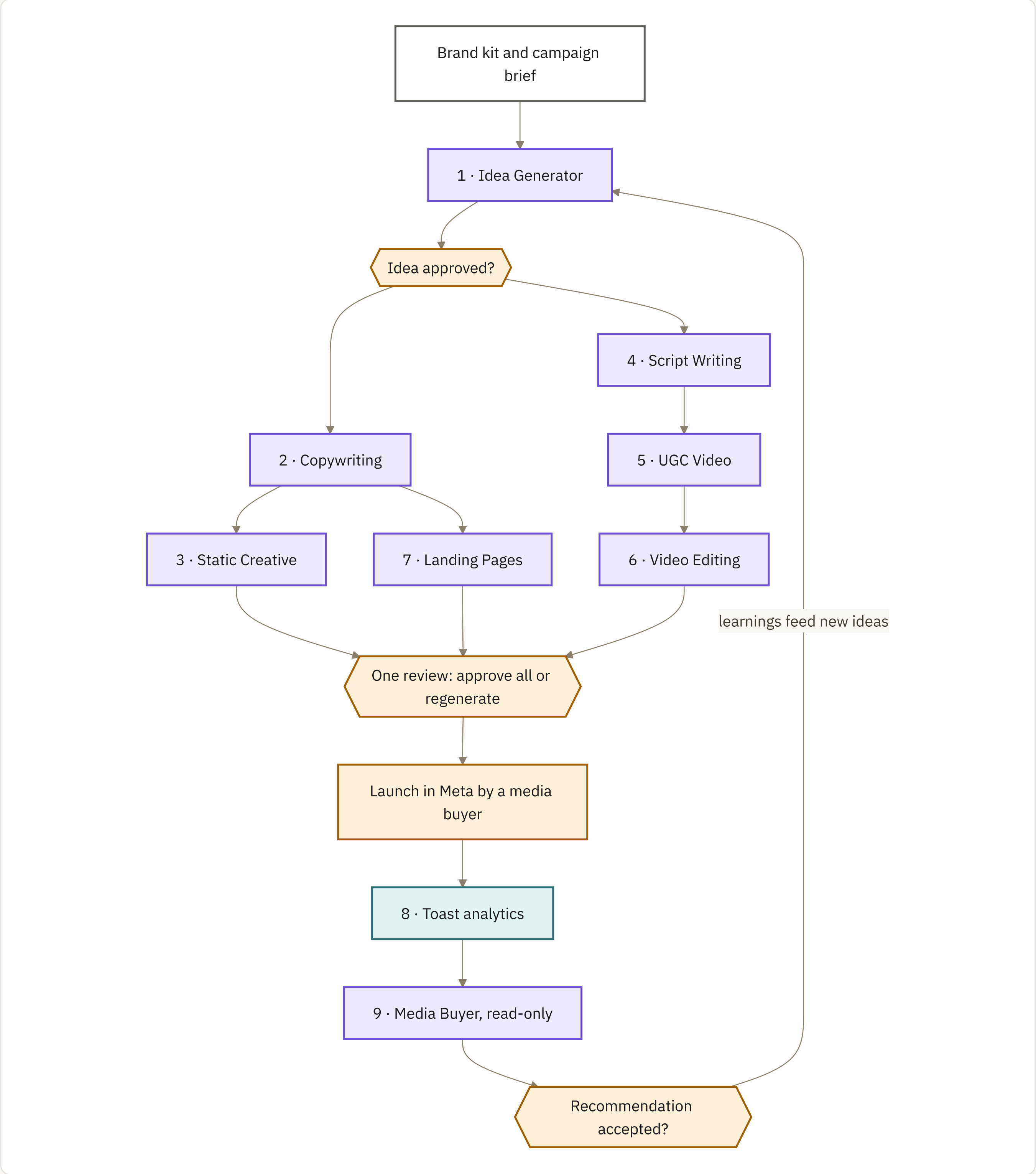
Every item from the brief's "What to prepare for the final interview", with a one-line answer. Click an item to jump to the detail.

✓	**Overall approach**	One connected system: brief in, nine agents, one human approval, results feed the next ideas. See: The ecosystem
✓	**Workflow for every area**	Input-to-output flow for all nine areas, showing AI steps, automatic checks and human decisions. See: Nine workflows
✓	**Tools / technology**	Claude for ideas and writing, OpenAI and Google for images, Whisper and video, HeyGen and ElevenLabs for presenters, Meta API for data, with the reason for each. See: Which AI we'd use for what
✓	**Demo / prototype**	A working web app: all nine agents and a one-click autopilot that builds a campaign package in about 35 seconds. See: The AI agents you can try today
✓	**AI involvement**	AI drafts and analyses. Code calculates. People approve anything that reaches a client or an ad account. See: Automate, assist or approve
✓	**Quality controls**	Facts-only rule, figure checking, banned phrases, schema validation, versioning, and checks that block approval. See: Quality controls
✓	**Maintenance**	Swap models per task, version prompts, test them against past briefs, and log every AI call. See: Keeping it healthy
✓	**Scale and cost**	Text AI costs pennies per brief; video is the main cost and is used selectively. Data is separated and costs are logged per client. See: Scale and cost
✓	**Milestones and timeline**	2 weeks, 30, 60 and 90 days, covering what comes first and why, dependencies, and how success is measured. See: Roadmap
✓	**Next steps**	What I'd build or improve first if I joined CerealBox. See: Next steps

END TO END

The ecosystem

Brief in, approved assets out, performance back in. Anything that leaves the system needs human approval, including ads, pages, client reports and budget changes.

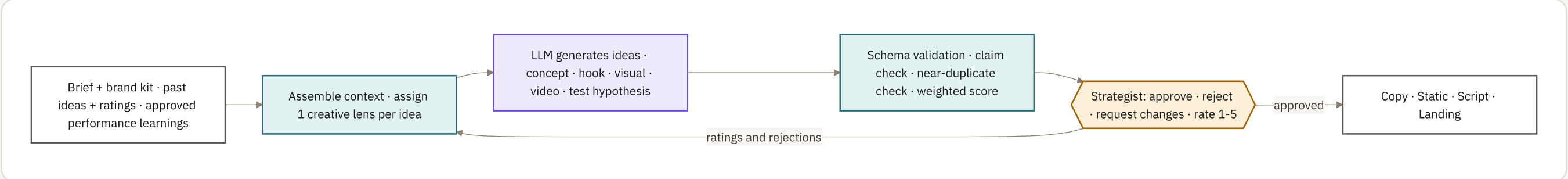

Brand kit and campaign brief
1 · Idea Generator
Idea approved?
4 · Script Writing
2 · Copywriting
5 · UGC Video
3 · Static Creative
7 · Landing Pages
6 · Video Editing
learnings feed new ideas
One review: approve all or regenerate
Launch in Meta by a media buyer
8 · Toast analytics
9 · Media Buyer, read-only
Recommendation accepted?

AREA BY AREA

Nine workflows

All nine run in the working demo on one shared provider layer, approval model and data structure. Seven are complete. UGC Video and Video Editing produce the full production plan; the paid video-rendering step plugs in later.

1 · AI Idea Generator WORKING DEMO


Brief + brand kit · past ideas + ratings · approved performance learnings
Assemble context · assign 1 creative lens per idea
LLM generates ideas · concept · hook · visual · video · test hypothesis
Schema validation · claim check · near-duplicate check · weighted score
Strategist: approve · reject · request changes · rate 1-5
approved
Copy · Static · Script · Landing
ratings and rejections

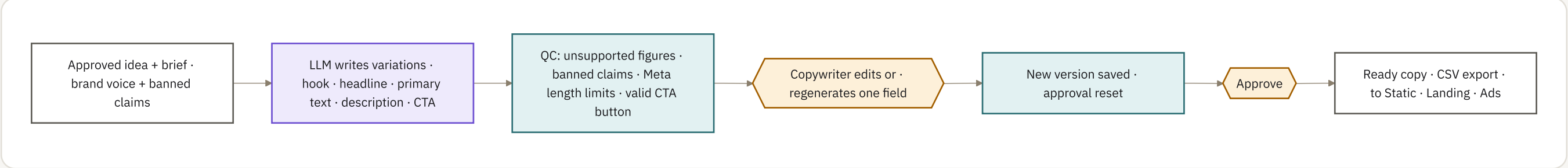
AI DOES
Generates 3–6 distinct concepts, each forced through a different lens (myth-buster, tangible impact, first-person POV and so on), and scores them against 8 criteria with a reason for each score.

HUMAN DOES
Approves, rejects with a reason, or rates each idea. Ratings and rejections steer the next batch.

QUALITY CONTROLS
Only facts listed in the brief's proof points may be used. Invented figures and banned claims are flagged. Scores help triage and are not presented as predictions.

PREFERRED TOOLS
Claude Opus 5.5 for idea generation (strong reasoning over long briefs, follows brand rules). **OpenAI GPT** as a second source of different ideas. Demo: Ollama Cloud (gpt-oss:120b).

2 · AI Copywriting WORKING DEMO


Approved idea + brief · brand voice + banned claims
LLM writes variations · hook · headline · primary text · description · CTA
QC: unsupported figures · banned claims · Meta length limits · valid CTA button
Copywriter edits or regenerates one field
New version saved · approval reset
Approve
Ready copy · CSV export · to Static · Landing · Ads

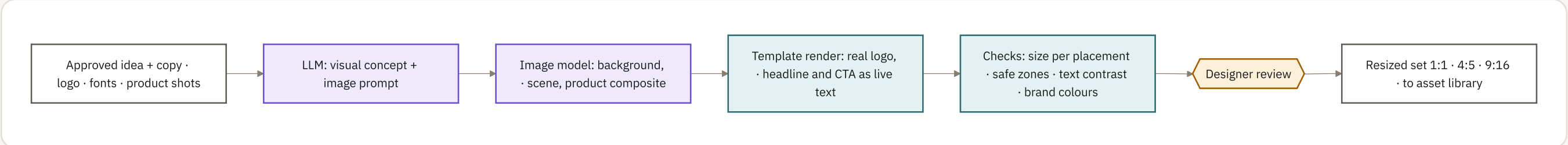
AI DOES
Writes 3–5 angled variations from the approved idea and can rewrite a single field on instruction, such as "make the hook shorter".

HUMAN DOES
Edits, compares versions and approves. Copy with a QC error such as a banned claim cannot be approved.

QUALITY CONTROLS
Every edit creates a new version and resets approval, so approved copy can't change silently. Each version records which provider and model produced it.

PREFERRED TOOLS
Claude Sonnet 5.5 for copy (natural tone, sticks to the facts given, good value at volume). **Claude Haiku 5.5** for bulk checks. Demo: Ollama Cloud.

3 · AI Static Creative WORKING DEMO


Approved idea + copy · logo · fonts · product shots
LLM: visual concept + image prompt
Image model: background, · scene, product composite
Template render: real logo, · headline and CTA as live text
Checks: size per placement · safe zones · text contrast · brand colours
Designer review
Resized set 1:1 · 4:5 · 9:16 · to asset library

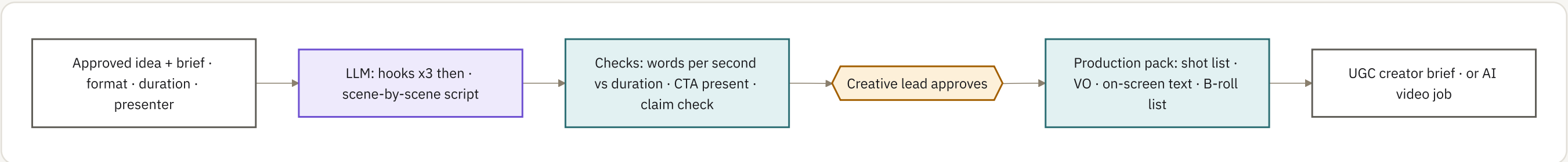
AI DOES
Turns the idea into a visual concept and generates imagery. It does not render the logo or headline text.

HUMAN DOES
A designer picks, adjusts and approves. Brand-critical pieces stay human-led.

QUALITY CONTROLS
Text and logo are placed by a template, never drawn by the image model, which avoids garbled lettering and off-brand logos. Each placement size is checked automatically.

PREFERRED TOOLS
OpenAI image model (gpt-image) or **Google Imagen** for scenes. **FLUX** (open, via ComfyUI or Hugging Face) for low cost. Claude writes the prompts; our template adds text. Demo: FLUX via Pollinations or Hugging Face.

4 · AI Script Writing WORKING DEMO


Approved idea + brief · format · duration · presenter
LLM: hooks x3 then · scene-by-scene script
Checks: words per second vs duration · CTA present · claim check
Creative lead approves
Production pack: shot list · VO · on-screen text · B-roll list
UGC creator brief · or AI video job

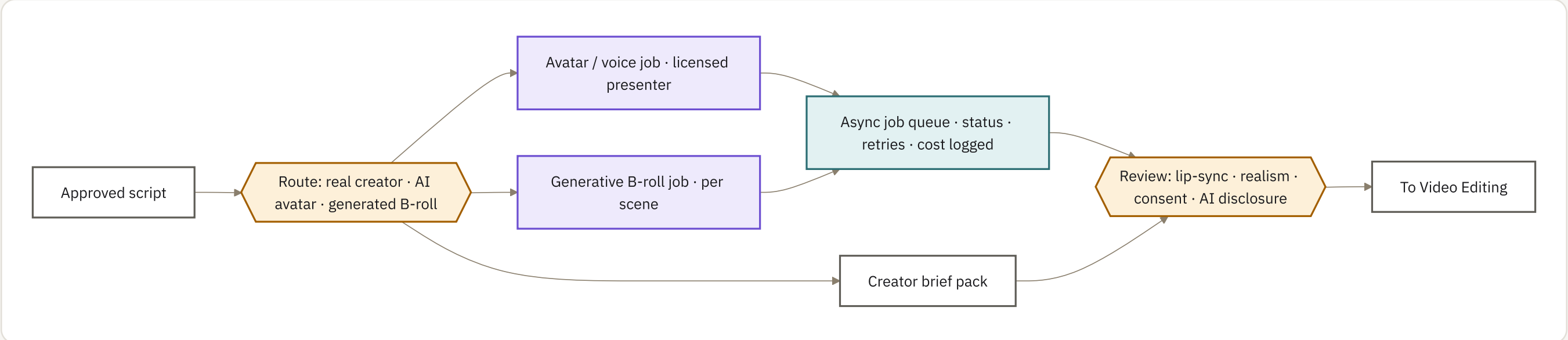
AI DOES
Writes UGC, testimonial-style, explainer and promo scripts as structured scenes: dialogue, VO, on-screen text and B-roll.

HUMAN DOES
Approves before any production money is spent.

QUALITY CONTROLS
Length is checked against the target duration at about 2.5 words per second. "Testimonial-style" scripts must never present invented customer stories as real.

PREFERRED TOOLS
Claude Sonnet 5.5 for scripts and scene structure; **OpenAI GPT** as an alternative. Demo: Ollama Cloud.

5 · AI UGC Video DEMO: PLAN ONLY


Approved script
Route: real creator · AI avatar · generated B-roll
Avatar / voice job · licensed presenter
Generative B-roll job · per scene
Async job queue · status · retries · cost logged
Creator brief pack
Review: lip-sync · realism · consent · AI disclosure
To Video Editing

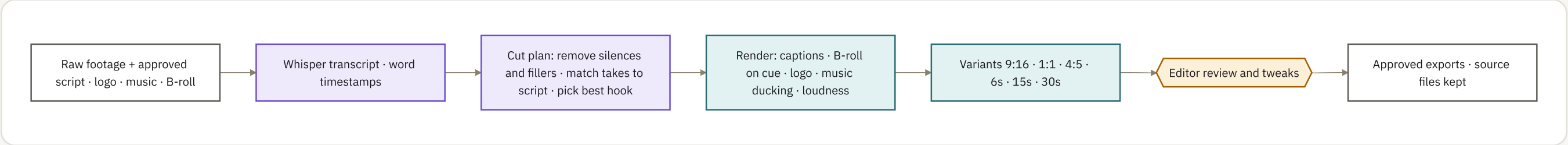
AI DOES
Produces avatar presenter takes and scene B-roll, with alternative takes per scene.

HUMAN DOES
Chooses the route per campaign and reviews every output. Fundraising appeals usually keep real people on camera.

QUALITY CONTROLS
Jobs only show as done when the provider confirms. Presenters must be licensed or have consented. AI content is disclosed in line with platform policy.

PREFERRED TOOLS
HeyGen for licensed avatars, **ElevenLabs** for voice, **Google Veo**, Runway or Kling for B-roll. Real creators for fundraising appeals. Demo: AI production pack and render job spec.

6 · AI Video Editing DEMO: PLAN ONLY


Raw footage + approved script · logo · music · B-roll
Whisper transcript · word timestamps
Cut plan: remove silences and fillers · match takes to script · pick best hook
Render: captions · B-roll on cue · logo · music ducking · loudness
Variants 9:16 · 1:1 · 4:5 · 6s · 15s · 30s
Editor review and tweaks
Approved exports · source files kept

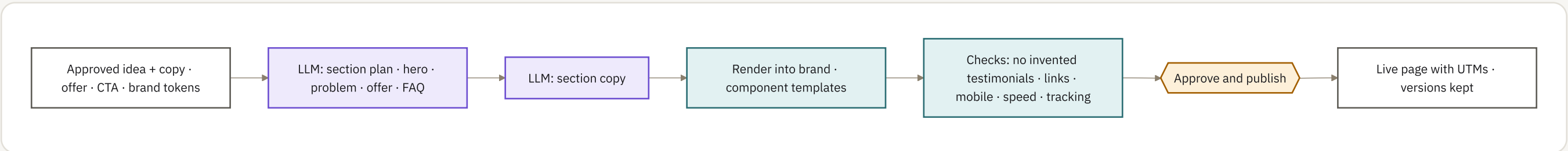
AI DOES
Transcribes, builds an edit decision list from the transcript and script, and places captions and B-roll.

HUMAN DOES
An editor adjusts pacing and approves. AI handles the first 80% of the work.

QUALITY CONTROLS
The pipeline checks for an audio track, resolution, caption sync and duration per placement. Originals are never overwritten.

PREFERRED TOOLS
OpenAI Whisper for word-timed transcripts, **FFmpeg** plus **Creatomate** or Shotstack for renders, Descript for editors. Claude writes the edit plan. Demo: AI edit plan.

7 · AI Landing Pages WORKING DEMO


Approved idea + copy · offer · CTA · brand tokens
LLM: section plan · hero · problem · offer · FAQ
LLM: section copy
Render into brand · component templates
Checks: no invented testimonials · links · mobile · speed · tracking
Approve and publish
Live page with UTMs · versions kept

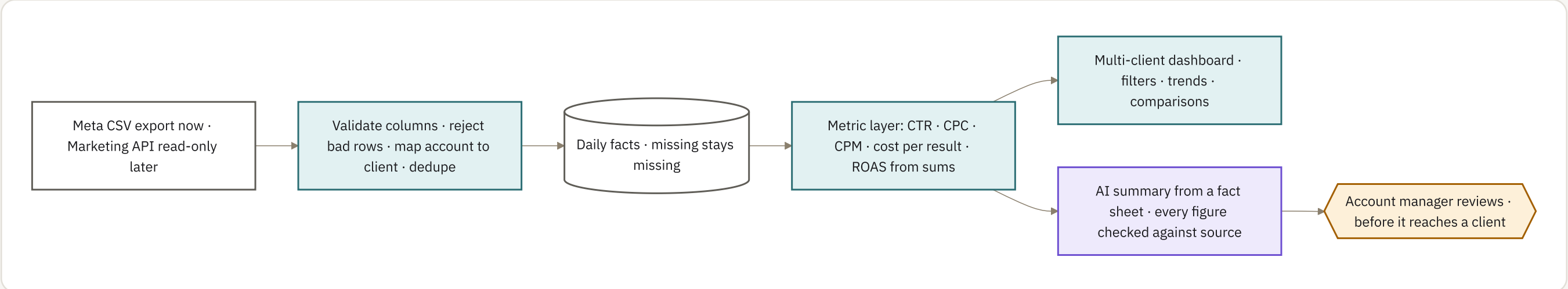
AI DOES
Plans the structure and writes each section, carrying over the ad's message so the ad and page match.

HUMAN DOES
Edits sections, previews desktop and mobile, and publishes.

QUALITY CONTROLS
Social proof sections appear only when real testimonials are supplied. CTA links are validated. A speed and mobile check runs before approval.

PREFERRED TOOLS
Claude Sonnet 5.5 for structure and copy, **Next.js on Vercel** (or Webflow/Unbounce) to publish, v0 for fast design drafts. Demo: page plan with desktop/mobile preview.

8 · Toast / Meta Ads Analysis WORKING DEMO


Meta CSV export now · Marketing API read-only later
Validate columns · reject bad rows · map account to client · dedupe
Daily facts · missing stays missing
Metric layer: CTR · CPC · CPM · cost per result · ROAS from sums
Multi-client dashboard · filters · trends · comparisons
AI summary from a fact sheet · every figure checked against source
Account manager reviews · before it reaches a client

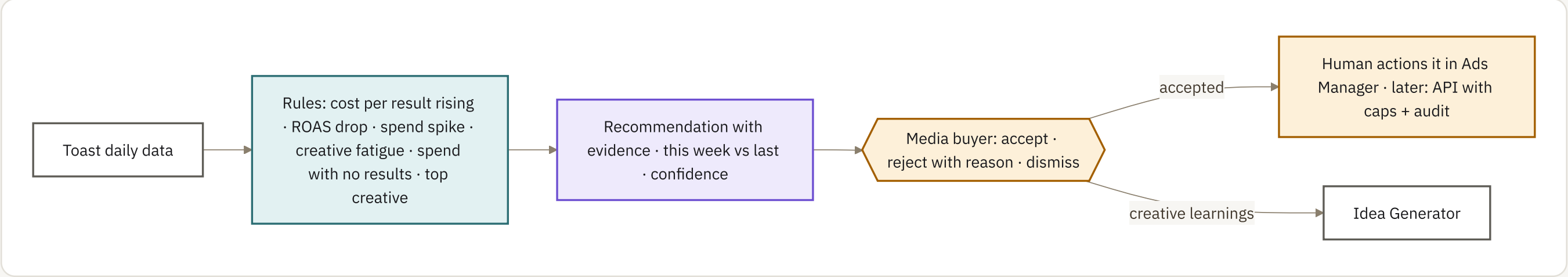
AI DOES
Writes plain-English summaries using only figures it is given. Any number not in the source data is flagged.

HUMAN DOES
Reviews summaries before they go to clients and maps new ad accounts to clients.

QUALITY CONTROLS
Blank cells stay "unavailable", never 0. Ratios are recomputed from totals. ROAS is shown only when conversion value is complete. Duplicate files and rows are skipped.

PREFERRED TOOLS
Meta Marketing API (read-only) on a nightly schedule (**n8n** or cron), Postgres or Turso for storage, **Claude Haiku 5.5** for figure-checked summaries. Demo: CSV import and dashboard.

9 · AI Media Buyer / Ads Agent WORKING DEMO


Toast daily data
Rules: cost per result rising · ROAS drop · spend spike · creative fatigue · spend with no results · top creative
Recommendation with evidence · this week vs last · confidence
Media buyer: accept · reject with reason · dismiss
accepted
Human actions it in Ads Manager · later: API with caps + audit
creative learnings
Idea Generator

AI DOES
Explains what changed and what to check, attaching the exact metrics that triggered each alert.

HUMAN DOES
Makes every decision. The agent is read-only and cannot pause, edit or publish anything.

QUALITY CONTROLS
Detection is rule-based and deterministic, so alerts are never invented. Thresholds are configurable. Every decision is audited.

PREFERRED TOOLS
A rules engine finds issues (no AI guessing). **Claude Sonnet 5.5** explains them with the numbers. Slack for alerts. Demo: rules plus accept/reject, with learnings feeding ideas.

TOOLS

Which AI we'd use for what

We pick the best tool for each job rather than one AI for everything. The app routes each task to a configured model, so a better or cheaper model can be swapped in without rebuilding anything.

JOB	PREFERRED	ALSO GOOD	WHY
Campaign ideas and strategy	**Claude Opus 5.5**	OpenAI GPT (flagship)	Strongest reasoning over long briefs; respects brand rules. A second model gives genuinely different ideas.
Ad copy, scripts, landing page copy	**Claude Sonnet 5.5**	OpenAI GPT	Natural, on-brand writing that sticks to the facts given; good value at volume.
Checks, tagging, quick summaries	**Claude Haiku 5.5**	OpenAI small models	Fast and low-cost for high-volume, simple jobs.
Image ads	**OpenAI gpt-image** or **Google Imagen**	FLUX (open; ComfyUI or Hugging Face)	Follows detailed prompts well. FLUX keeps costs low and can run privately. Text and logo always come from our template.
UGC presenter video	**HeyGen** + **ElevenLabs**	Google Veo, Runway, Kling (B-roll)	Licensed avatars and natural voices; real creators stay first choice for fundraising.
Transcription and editing	**OpenAI Whisper** + **FFmpeg**	Creatomate or Shotstack, Descript	Accurate word timings for captions and cuts; automated renders in every size.
Landing pages	**Claude Sonnet 5.5** + **Next.js on Vercel**	Webflow, Unbounce, v0	Page matches the ad; published only after approval.
Ads data and reporting	**Meta Marketing API** (read-only)	n8n, Looker Studio	Numbers computed by code, not AI; AI only explains them.
Automation and agents	**Our app** + **n8n**	Claude Agent SDK	Core logic in code we control; visual automations where they help.

`In the demo, all text agents run on Ollama Cloud (gpt-oss:120b) because it has a free tier. Claude and OpenAI models plug in through the same provider layer (OpenRouter adapter today, direct adapters are one file each). Check current model versions and pricing before committing.`

WORKING DEMO

The AI agents you can try today

A working web app (Next.js, deployable to Vercel). Describe a campaign once; the AI autopilot runs every agent in about 35 seconds, then a person approves the package in one place.

AGENT	IN THE DEMO	STATUS
Idea Generator	3–6 ideas from different angles, scored with reasons, picks the strongest; learns from ratings and ad results	Working
Copywriting	2–5 ad versions with fact, banned-phrase and Meta length checks; edit or rewrite one field; CSV export	Working
Static Creative	Concepts, real AI images and finished PNG ads in 1:1, 4:5 and 9:16, text set by template	Working
Script Writing	Hooks and scene-by-scene scripts timed to 15–60 seconds, CTA check	Working
UGC Video	Presenter brief, shot list, consent and AI-disclosure notes, render job spec	Plan only
Video Editing	Timeline, captions, B-roll cues, music and export sizes	Plan only
Landing Pages	Section plan and copy with desktop/mobile preview; never publishes on its own	Working
Toast (Meta analysis)	CSV import with validation, multi-client dashboard, AI summary with every figure checked	Working
Media Buyer	Flags rising costs, tired ads, spend spikes, wasted spend and winners; read-only; accepted lessons feed new ideas	Working

Across every agent: Approve · Regenerate (with a note) · Reject, "Fill with AI" and per-field suggestions on every form, AI settings per campaign, and a full log of decisions and AI calls.

JUDGEMENT

Automate, assist or approve

TASK	MODE	WHY
Data ingest, validation, metric calculation, alerts	Automate	Deterministic, testable, and low risk if it fails visibly
Ideas, copy, scripts, page drafts, edit plans, summaries	AI assists	AI drafts are fast and cheap. Quality varies, so a human chooses
Anything a client or the public sees	Human approves	Brand, legal and fundraising-regulation risk
Budget, bid, pause or publish in ad accounts	Human approves	Money moves. Read-only now; API writes later only with caps and audit

QUALITY CONTROLS

How we reduce bad outputs, errors and hallucinations

Facts only from the brief
The AI may only use figures and claims listed as proof points. Any other number is flagged, and in summaries checked against the real data.

Brand rules enforced
Each client's banned phrases block approval until fixed. Absolute claims ("guaranteed", "best", "100%") and invented testimonials are flagged.

Structured, validated output
Every AI answer must match a strict format. Invalid answers are retried once, then fail visibly; mock output is never passed off as AI.

Different, not repetitive
Ideas use different creative angles, near-duplicates of past ideas are flagged, and reviewer ratings steer the next batch.

Platform-ready
Meta length limits, valid buttons, exact ad sizes and Stories safe zones are checked automatically. Text on ads comes from a template, never from the image model.

Nothing changes silently
Edits and regenerations create new versions and reset approval. Every decision and AI call is logged.

Data you can trust
Bad and duplicate rows are rejected, missing values stay "unavailable" rather than zero, and ratios are calculated from totals.

Humans hold the keys
The Media Buyer is read-only. Nothing is published, launched or changed in an ad account without a person.

ROADMAP

2 weeks, 30, 60 and 90 days

Text workflows and data come first because they are cheap, measurable and feed everything else. Video comes last because it is the most expensive, the least mature and carries the most brand risk.

MILESTONE	BUILD	DEPENDS ON	DONE WHEN
Weeks 1–2	Foundation and pilot on real briefs. *Already prototyped:* all nine agents, one-click autopilot, approvals, Toast and Media Buyer	Brand kits and 2–3 real briefs, sample exports	3 pilot briefs run end to end. At least 50% of ideas rated 4/5 or higher by strategists. Toast totals match Ads Manager within 1%
Day 30	Meta Marketing API read-only ingest for all clients. Slack alerts. Script Writing. Static Creative v1 (AI imagery and templates). Postgres, logins and roles	Meta app review, access to ad accounts, design templates	Every account refreshes nightly. At least 70% of alerts marked "useful". First copy draft in under 10 minutes
Day 60	Video Editing pipeline (Whisper and FFmpeg or Creatomate). Landing Pages. UGC pilot with one avatar provider plus creator brief packs. Client report export	Footage library, provider trials, hosting for pages	Short-form edit time down 50%. 2 landing pages live through approval. First UGC tests running
Day 90	All 9 workflows connected. Learning loop on. Cost and usage dashboard. Prompt regression tests. Optional Media Buyer write actions behind caps	Pilot feedback, a decision on write permissions	All 9 used on at least 2 live clients. Cost per asset tracked. Quality scores stable across model changes

Keeping it healthy

- Each task has its own model setting, so a model can be swapped without code changes
- Prompts are versioned and checked against a fixed set of past briefs before any model switch
- Every AI call is logged with provider, latency, tokens and failures
- Reviewer ratings and rejections are the quality metric to track week by week
- Rule thresholds live in config and are tuned from "useful / not useful" feedback

Scale and cost

- Text AI (ideas, copy, summaries) costs pennies per brief on small models. Free tiers cover a pilot
- Images: a few pence each on hosted APIs, or close to zero on a local GPU
- Avatar and generative video is the main cost, usually paid per minute or by subscription, so it is used selectively
- Meta API access is free but rate-limited, handled with a nightly batch pull
- Every client's data is kept separate. Costs are logged per client and campaign

`Prices are indicative. Check current vendor pricing before committing.`

NEXT STEPS

If I joined CerealBox, what I'd do next

1. **Week 1: learn the real work.** Sit with strategists, copywriters and media buyers. Collect 10–20 past briefs and their best-performing ads to use as a test set for every AI change.
2. **Plug in the real data.** Connect the Meta Marketing API (read-only) so Toast refreshes every client nightly, and send Media Buyer alerts to Slack.
3. **Upgrade the writing agents.** Switch ideas and copy to Claude, compare the results side by side with the team on the test set, and keep whichever wins.
4. **Brand kits per client.** Load logos, fonts and colours so image ads and landing pages come out fully on-brand.
5. **Connect video production.** Add the render step for UGC (HeyGen and ElevenLabs) and for editing (Whisper and FFmpeg), starting with one pilot client.
6. **Make it team-ready.** Logins and roles, a hosted database, and a weekly client report built from Toast.
7. **Measure and improve.** Track hours saved, approval rate and edits needed for each agent, then improve the weakest agent each month.

Prepared by Sohaib Ahmed for the CerealBox Agency AI automation challenge. All client names and data in the demo are fictional sample data. No live CerealBox systems or ad accounts were accessed.